ECE 20001: Electrical Engineering Fundamentals I – Spring 2026

CRN 26675

General Class Information

- Course Number: ECE 20001
- Course Name: Electrical Engineering Fundamentals I
- Course Reference Number (CRN): 26675

General Description

This course covers fundamental concepts and applications for electrical and computer engineers as well as for engineers who need to gain a broad understanding of these disciplines. The course starts by the basic concepts of charge, current, and voltage as well as their expressions with regards to resistors and resistive circuits. Essential concepts, devices, theorems, and applications of direct-current (DC), 1st order, and alternating-current (AC) circuits are subsequently discussed. Besides electrical devices and circuits, basic electronic components including diodes and transistors as well as their primary applications are also discussed.

Learning Outcomes

1. An ability to analyze linear resistive circuits.

2. An ability to analyze 1st order linear circuits with sources and/or passive elements.

3. An ability to analyze electronic circuits with diodes and transistors.

Learning Resources

The class will be primarily based on class notes in the form of an e-textbook that will include PDF files and videos that will be made available through the class's website.

Assignments/Assessments

Typically, one homework assignment will be distributed every week. There will be two exams during the semester and one final exam.

Grading Scale

The final class grade will be calculated from the weighted average of the homework and exam scores.